Universidade de Brasília
Faculdade do Gama

Digital Integrated Circuit Design

Laboratory 2 – CMOS Inverter

Objectives:

To carry out the following tasks:

1. Implement the schematic of a CMOS inverter;
2. Simulate and analyze the inverter's DC transfer characteristic by performing a parametric sweep to optimize the dimensions (W/L) of the PMOS and NMOS transistors;
3. Perform a transient analysis to observe and evaluate how transistor sizing affects the inverter's rise and fall times.

1. Schematic

In your working directory, create a new folder named **Lab2**. Launch **Cadence** and create a new library named **inversor**. Within the **inversor** library, create a new schematic cell named **inv**, representing a CMOS inverter, as shown in **Figure 1**. Add the input and output pins using the menu: **Create → Pin**. For this initial implementation, keep the default **W (width)** and **L (length)** values for both the **pmos1v** and **nmos1v** transistors.

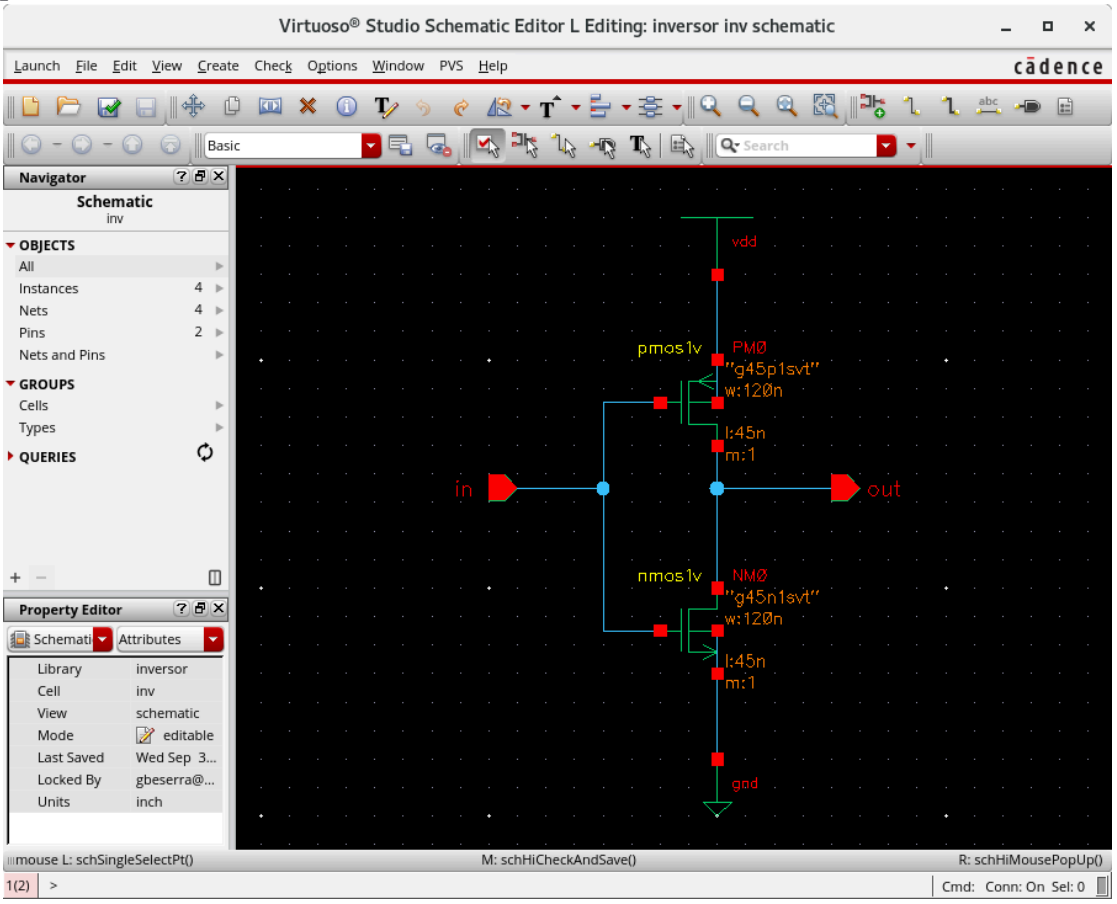


Figure 1

After completing the schematic, run **Check and Save** to validate and save the design. Next, create a symbol for the inverter using the following menu path: **Create → Cellview → From Cellview**. You may keep the automatically generated symbol as shown in **Figure 2**, or edit it to visually represent a standard inverter symbol, as illustrated in **Figure 3**.

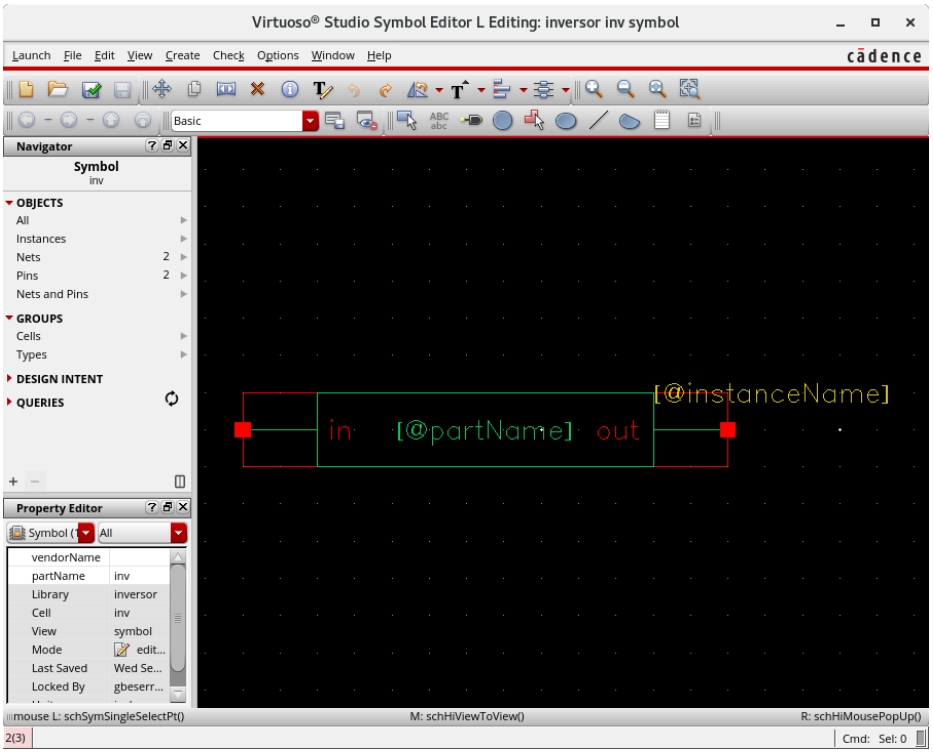


Figure 2

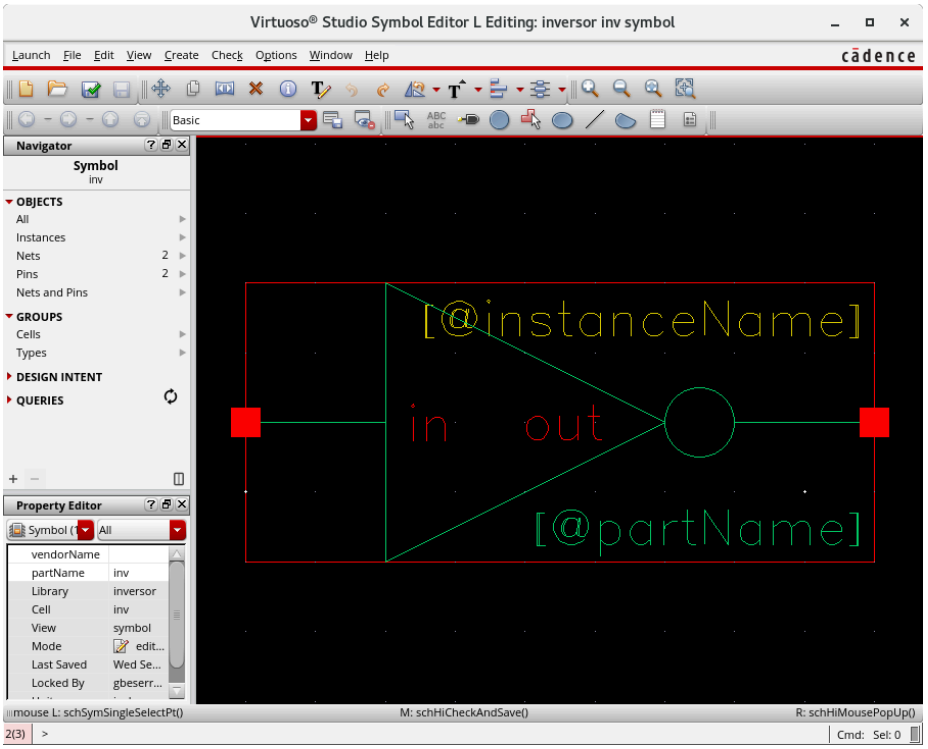


Figure 3

2. DC Curve

The simulation can be performed by creating a testbench. To do this, create a new schematic named **inv_testbench** in the **inversor** library and place the components as shown in **Figure 4**. First, instantiate the **symbol view** of the inverter. Then, place a **vpulse** voltage source (from **analogLib**) at the input of the inverter and a **5 fF capacitive load** (component **cap** from **analogLib**) at the output. Bring up the properties of **vpulse**. Change **Voltage 2** to **1.0 V**. Use **50 ps** for both rise and fall times, and use **5 ns** and **10 n**s for the pulse width and period, respectively (see **Figure 5**).

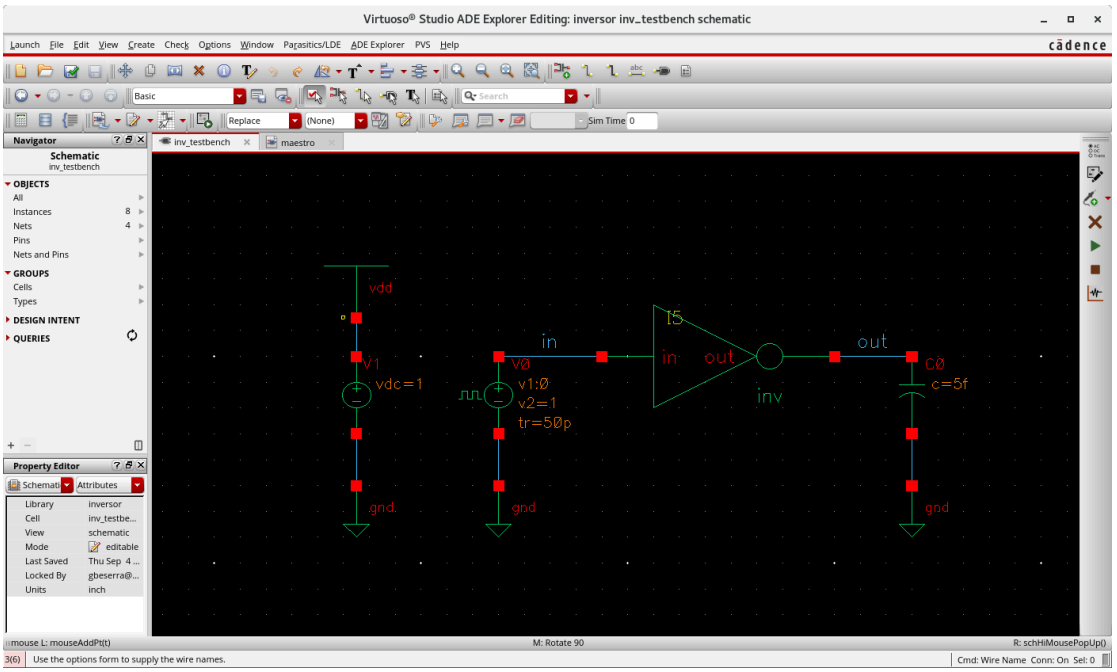


Figure 4

Set the **reference voltage** to **1 V**, as shown on the left side of **Figure 4**. Next, right-click on the inverter symbol and choose **Descend Edit** (or press **Shift + X**). In the schematic view of the inverter, select the **pmos1v** properties and change the **Finger Width** to **w** and **Fingers** to **1**. Click **Check and Save** to validate the modifications, and press **Shift + B** to return to the **inv_testbench** schematic.

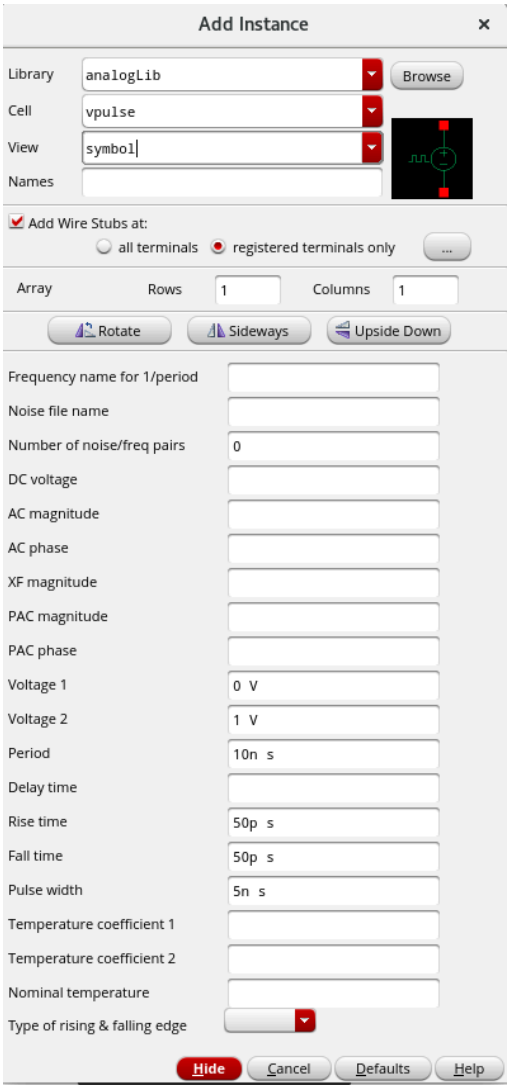


Figure 4

Open the **Analog Design Environment (ADE Explorer)** and create a new **maestro view** for your testbench. To set up the DC analysis, go to **Analyses → Choose**, then select **dc**. In the ***Sweep Variable List***, choose **Component Parameter** and click on the **vpulse** voltage source connected to the inverter input. In the window that appears, select **dc** and click **OK.** Back in the previous window, configure the ***Sweep Range*** by setting **Start = 0** and **Stop = 1** (**Sweep Type**: Automatic), then click **OK**. This configuration sets up a DC simulation sweep in which the simulator varies the input voltage from 0 V to 1 V and computes the corresponding DC voltages at all circuit nets for each input value within that range.

Select ***Outputs → To be plotted → Select on Design*** in the **ADE Explorer** window. Then, click on the inverter output wire. Next, in the ***Variables*** menu, select ***Copy from Cellview***. The variable ***w*** should appear in the ***Design Variables*** list.

The optimal PMOS-to-NMOS sizing ratio can be determined through a **parametric simulation**, where a simulation for each value of the PMOS width in the sweep range is performed. The goal is to observe how variations in that parameter affect the circuit's performance, enabling more informed design decisions. In the **Design Variables** list, locate the variable **w**, then double-click the blank field next to it. Click the button with the three dots (**...**) to open the **Parameterize** window. In the **Add Specification** menu, select **From/To**, and fill in the following values:

From: 120n
To: 270n
Total Steps: 7
Step Mode: Auto

This setup will sweep the width **(w)** parameter from **120 nm to 270 nm**, using 7 evenly spaced values within that range. Once the simulation is complete, a plot window will automatically open, displaying the output response for each value of the swept width. Maximize the plot window to better view the results. On the left side, a legend displays the color corresponding to each waveform, along with the width **(w)** value used to generate that specific curve. You can click on any entry in the legend to highlight the associated waveform in the plot. Go to **Marker → Create Marker**, then select **Point**. Set the position to **X = Y = 0.5 V** (Vdd/2). You will notice that one of the **Vout** curves intersects near that value. By checking the legend, you can identify the corresponding width value for this curve. Once this value is determined, you can size the PMOS transistor to have approximately the same effective resistance as the NMOS transistor. This ensures balanced switching characteristics, leading to more symmetrical rise and fall times in the inverter's output.

3. Transient Analysis:

Open the ***inv_testbench*** schematic and assign the value found in the previous simulation for ***Wp***. Open ***ADE Explorer*** and set up the transient analysis:

- **Analyses -> Choose**: tran
- **Stop Time**: 20n

In the **ADE Explorer** window, select ***Outputs → To be plotted → Select on Design***. Then, click on the wires for the output and input of the inverter and run the simulation. The waveforms shown in Figure 5 will appear.

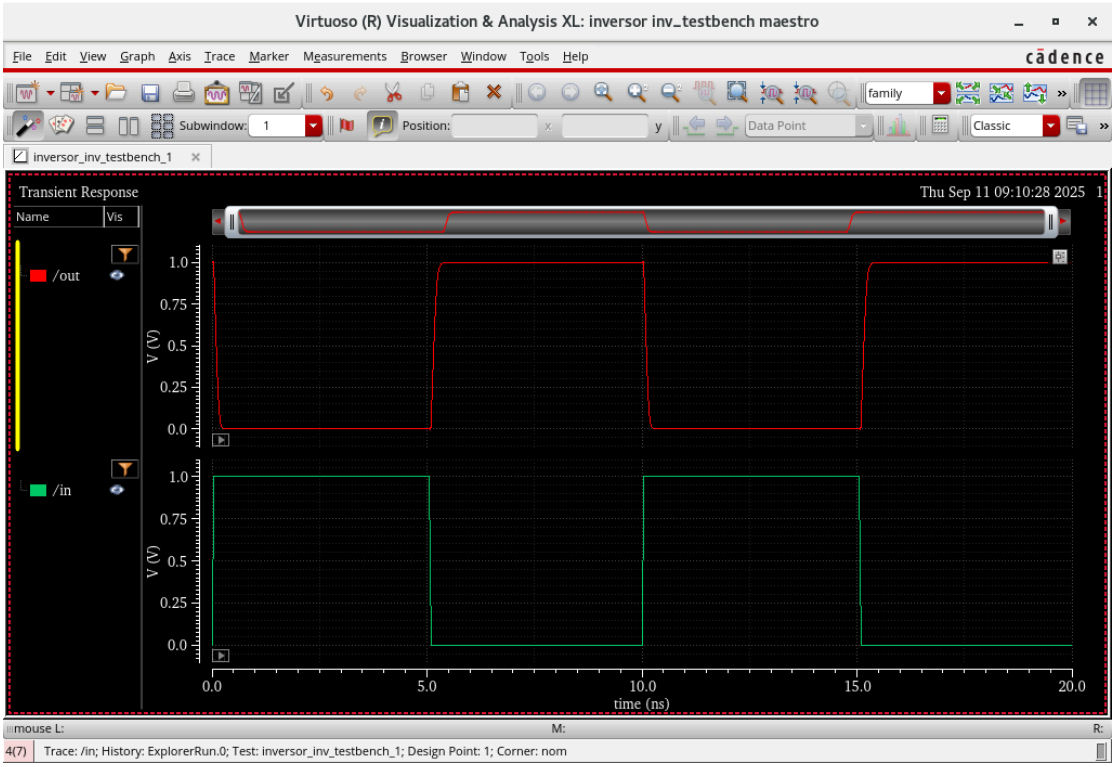


Figure 5

Using the calculator functions (***Tools*** → ***Calculator***), determine the following parameters:

t_r =
t_f =
t_{pHL}=
t_{pLH} =
t_p =

Vary ***Wp*** from **120 nm** to **270 nm**, as shown in the table below, and observe the effect of sizing on the inverter output.

Wp (nm)	Tf (ps)	Tr (ps)
120		
145		
170		
195		
220		
245		
270		

Observe for which ***Wp*** the rise and fall times are equal (or have the smallest difference) and compare with the value obtained in the DC simulation.

5. Tasks

1. Screenshot showing the DC simulation, indicating the value of ***w*** found, along with a brief textual description.
2. Screenshots showing the determination of the parameters for the transient analysis. Fill in the parameter values and the table.